

Roll No.

TMA-403

B. TECH. (PE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022
NUMERICAL AND GEOSTATISTICAL METHODS

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.
1. (a) The median and mode of the following wage distribution are ₹ 33.5 and ₹ 34 respectively. However, three frequencies are missing. Determine their values : (CO1)

Wages (in hundred ₹)	Frequencies
0—10	4
10—20	16
20—30	?
30—40	?
40—50	?
50—60	6
60—70	4
Total	230

P. T. O.

- (b) An analysis of production rejects resulted in the following figures :

(CO1)

No. of rejects per operator	No. of operators
21—25	5
26—30	15
31—35	28
36—40	42
41—45	15
46—50	12
51—55	3

Calculate the coefficient of skewness and kurtosis and comment on the results.

- (c) From the following frequency distribution of heights of 360 boys in the age group 15-20 years, calculate the standard deviation and variance :

(CO1)

Height (in cms.)	No. of boys
126—130	31
131—135	44
136—140	48
141—145	51
146—150	60
151—155	55
156—160	43
161—165	28

2. (a) A company gives on the job training to its salesman which is followed by a test. It is considering whether it should terminate the services of any salesman who does not do well in the test. The following data give the test scores and sales made by nine salesman during the last one year :
(CO2)

Test Scores	Sales (in Thousands)
14	31
19	36
24	48
21	37
26	50
22	45
15	33
20	41
19	39

- (b) A panel of two judges, A and B, graded seven T. V. serial performances by awarding marks independently as shown in the following table :
(CO2)

Performance	Marks by A	Marks by B
1	46	40
2	42	38
3	44	36
4	40	35
5	43	39
6	41	37
7	45	41

The eighth T. V. performance which judge B could not attend, was awarded 37 marks by judge A. If the judge B had also been present, how many marks would be expected to have been awarded by him to the eighth T. V. performance ?

(c) From the following data obtain the two regression equations : (CO2)

X	Y
91	71
97	75
108	69
121	97
67	70
124	91
51	39
73	61
111	80
57	47

3. (a) Two research laboratories have independently produced drugs that provide relief to arthritis patients. The first drug was tested on a group of 100 arthritis patients and produced an average of 8.5 hours of relief with a standard deviation of 2 hours. The second drug was tested on 75 patients, producing an average of 7.8 hours of relief with a standard deviation of 1.5 hour. At a significance level of one percent, does the first drug provide a significantly longer period of relief ? (CO3)

(5)

- (b) The following table gives the number of good and bad parts produced by each of three shifts in a factory : (CO3)

Shift	Good	Bad
Day	900	130
Evening	700	170
Night	400	200

Is there any association between the shift and the equality of parts produced ? Chi-Square tabulate value is 5.991.

- (c) The following data represent the yield in quintals of Indian corn on ten sub-divisions of equal areas of two agricultural plots in which plot I was a control plot treated the same as plot II, except for the amount of phosphorous applied as a fertilizer. (CO3)

Plot I	Plot II
6.2	5.6
5.7	5.9
6.5	5.6
6.0	5.7
6.3	5.8
5.8	5.7
5.7	6.0
6.0	5.5
6.0	5.7
5.8	5.5

P. T. O.

Is there a significant difference between the yields on the two plots, using the difference between their means by a criterion of judgement. The value of t for 18 degrees of freedom at 5% level of significance is 0.0072.

4. (a) Explain normal distribution. How is it important in geological sequences? Write the formula for normal distribution. Draw a figure of normal distribution with variance. (CO4)
- (b) Explain the characteristic of Unimodal, Bimodal, left-skewed, right skewed, symmetric distributions with figure. How are these models important in geological data analysis? (CO5)
- (c) What is forecasting? What are general characteristics of forecasts? How many types of forecasting methods are available? (CO4)
5. (a) Explain Interpolation, Extrapolation, Global interpolation with figures. What is inverse distance weighted technique? Explain with example. (CO4)
- (b) Explain Semi variance. Write the formula for semi variance. What is the significance of sill, nugget in a variogram analysis? (CO5)
- (c) What is Geostatistics and how does it differ from statistic? What is feature space and Geographic space? Explain the bubble map with example. (CO4)

Roll No.

TPE-401

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

PETROLEUM GEOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe paraffins, cycloparaffins and aromatic. (CO1)
(b) Describe biogenic theory of origin of hydrocarbons. (CO1)
(c) Describe abiogenic theory of origin of hydrocarbons. (CO1)
2. (a) What is migration ? Describe primary migration. (CO3)
(b) What is source rock ? Describe source rocks in detail. (CO3)
(c) Describe stratigraphic traps in detail. (CO3)
3. (a) What is reservoir rock ? Describe in detail. (CO4)

P. T. O.

(2)

- (b) What are petroleum traps ? Describe structural traps in detail. (CO4)
- (c) Describe surface and subsurface occurrence of hydrocarbon. (CO1)
- 4. (a) Describe dew point, pour point, flash point and burning point of oil in detail. (CO1)
- (b) Describe hydrocarbon resource, inplace hydrocarbon, ultimate reserves and reserves. Also write formula for estimation of inplace oil, solution gas and ultimate reserves. (CO5)
- (c) How are structure contour maps prepared and what are their uses ? (CO5)
- 5. (a) Describe SPE classification and former Soviet Union classification of reserves with definition of each category. (CO6)
- (b) Describe ultimate recovery of reserves with a suitable diagram. (CO6)
- (c) Describe colour, optical activity, API gravity and viscosity of oil in detail. (CO2)

Roll No.

TPE-402

**B. TECH. (PE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

RESERVOIR ENGINEERING-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Describe the following factors : (CO1, CO5, CO6)
- (i) Wettability
 - (ii) Critical oil saturation
 - (iii) Absolute permeability
 - (iv) Effective permeability
- (b) With the help of a diagram describe original porosity, induced porosity and factors affecting porosity. (CO1, CO5, CO6)
- (c) Explain the following reservoirs near critical gas condensate, retrograde gas condensate and dry gas reservoir. (CO1, CO5, CO6)

P. T. O.

2. (a) With the help of a table clearly explain the concept of grain size classification of clastic sediments. (CO1, CO2, CO5)
- (b) Define the following : (CO1, CO2, CO5)
- (i) Total porosity
 - (ii) Effective porosity
 - (iii) Microporosity
 - (iv) Macroporosity
 - (v) Mesoporosity
 - (vi) Texture
 - (vii) Framework grains
 - (viii) Matrix
 - (ix) Sweet crude
 - (x) Sour crude
- (c) Explain any *three* characteristics of principal classes of crude oils. (CO1, CO2, CO5)
3. (a) Construct a diagram and explain radial flow and linear flow. (CO5, CO6)
- (b) Explain the following : (CO5, CO6)
- (i) Laminar Flow
 - (ii) Turbulent Flow
 - (iii) Steady Flow
 - (iv) Unsteady Flow
 - (v) Uniform and Non-Uniform Flow
 - (vi) Ideal Fluid

(vii) Real Fluid

(viii) Rotational Flow

(ix) Irrotational Flow

(c) Describe the concept of WOR, GOR, gas coning and water coning.

(CO5, CO6)

4. (a) An oil reservoir exists at its bubble-point pressure of 3000 psia and temperature of 160° F. The oil has an API gravity of 42° and gas-oil ratio of 600 scf/STB. The specific gravity of the solution gas is 0.65. The following additional data are also available : (CO4, CO5)

- Reservoir area = 640 acres
- Average thickness = 10 ft
- Connate water saturation = 0.25
- Effective porosity = 15%

Calculate : Specific Gravity Y_o , Initial Oil Formation Volume Factor, Pore Volume, Initial Oil in Place.

- (b) (i) Calculate the arithmetic average and thickness-weighted average from the following measurements :

Sample	Thickness, ft.	Porosity, %
1	1.0	10
2	1.5	12
3	1.0	11
4	2.0	13
5	2.1	14
6	1.1	10

P. T. O.

- (ii) Calculate average oil and connate water saturation from the following measurements : (CO4, CO5)

Sample, %	hi, ft	Porosity, %	So, %	Swc, %
1	1.0	10	75	25
2	1.5	12	77	23
3	1.0	11	79	21
4	2.0	13	74	26
5	2.1	14	78	22
6	1.1	10	75	22

- (c) (i) Describe compressibility, types of compressibility and capillary pressure.

- (ii) Calculate the pressure difference, i.e., capillary pressure, and capillary rise in an oil-water system from the following data :

(CO4, CO5)

$q = 30^\circ$, $r_w = 1.0 \text{ gm/cm}^3$, $r_o = 0.75 \text{ gm/cm}^3$, $r = 10^{-4} \text{ cm}$, $\sigma_{ow} = 25 \text{ dynes/cm}$

5. (a) Describe solution gas drive and gas cap drive. (CO1, CO2, CO3)

- (b) With the help of a diagram explain gravity drainage drive and combination drive. (CO1, CO2, CO3)

- (c) Write a short note on water influx on gas condensate.

(CO1, CO2, CO3)

Roll No.

TPE-403

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

HEAT AND MASS TRANSFER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Fourier's law of conduction and hence how the conductivities vary with respect to temperature in solid metals, liquid and gases ?

Solve : Two slabs each 240 mm thick have thermal conductivities of 14.5 W/m°C and 210 W/m°C. If the temperature of the face of hot surface is 220°C and the outside surface of the other slab is 30°C, determine the heat flow rate per unit area through the composite system and the intermediate temperature. (CO1, CO2)

P. T. O.

- (b) Define insulation. What are the applications of insulation ? Derive an expression for critical radius of insulation for a spherical pipe. What is the ratio of critical radius of insulation of sphere and cylinder ?

(CO1, CO2)

- (c) Define heat transfer coefficient and hence the overall heat transfer coefficient. Derive an expression for the overall heat transfer coefficient considering the convective and conductive heat transfer over the solid metal boundary.

(CO3)

2. (a) Discuss free convection from a flat surface for a vertical plate, hot or cold suggested by Churchill and Chu. Is this equation applicable for an inclined plane ? If yes, then what should be the minimum angle of inclination ? If no, why ?

(CO3, CO4)

- (b) Air flows over a flat surface, 2 m in length oriented in the direction of flow and of sufficient breadth, maintained at 150°C. The pressure is 1 atm. and the bulk air temperature is 30°C. If the air velocity is 12 m/s, determine (i) the local heat transfer coefficient as a function of longitudinal position, (ii) the average heat transfer coefficient and (iii) the rate of heat loss from the surface.

(CO1, CO4)

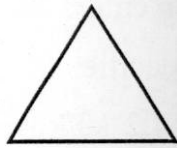
The relevant physical properties of air are taken as :

$$\rho = 0.962 \text{ kg/m}^3, \mu = 2.131 \times 10^{-5} \text{ kg/m-s}, k = 0.031 \text{ W/m}^\circ\text{C}, \\ C_p = 1.01 \text{ kJ/kg } ^\circ\text{C}.$$

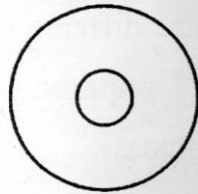
- (c) Discuss free convection with a suitable correlation from a cylinder in terms of Rayleigh's number.

(CO1, CO2)

3. (a) Define Black and Gray Body. What are the laws and relations for radiation for surfaces on the basis of the concept of a black body ? Hence deduce the expression for Kirchhoff's law (i.e., $\varepsilon = \alpha$). (CO1)
- (b) Define view factor. How many view factors involved, known and number of view factor that can be determined if we have 4-surface enclosure ? Calculate the view factor for the following configurations shown in the figure : (CO1, CO2)
- (i) Equilateral triangle :



- (ii) Sphere enclosed in a sphere :



- (c) Define radiation intensity and the spectral intensity with its unit. Derive the expression for the spectral emissive power of a black body over a hemisphere. (CO2, CO5)
4. (a) What are the types of heat exchanger on the basis of nature of (i) heat exchange process, relative, (ii) relative direction of fluid motion and (iii) design and constructional features ? Discuss LMTD of counter and concurrent flow in a heat exchanger. (CO1, CO5)
- (b) Write the assumptions made during the design aspects of heat exchanger. Derive the relation of logarithmic mean temperature difference [LMTD] for counter flow heat exchanger. Specify the symbol used. (CO4, CO5)

P. T. O.

(c) Illustrate the following :

(CO3, CO6)

- (i) LMTD
- (ii) Effectiveness
- (iii) Capacity ratio
- (iv) NTU

Hence solve : A crude oil with a flow rate of 1000 kg/hr. is to be cooled using water in a double-pipe counter-flow heat exchanger from a temperature of 80°C to 40°C. The water enters the exchanger at 20°C and leaves at 40°C. The specific heat capacities of the oil and the water at constant pressure are 2 kJ/kg-K and 4.2 kJ/kg-K, respectively. The overall heat transfer coefficient is 0.25 kW/m² K. Neglecting the heat loss and using the log mean temperature difference (LMTD) method, the minimum heat exchanger area (m²) required for the operation is (rounded off to two decimal places).

5. (a) Illustrate the modes of Mass Transfer. Define the mass transfer terms with the mathematical expression of :

(CO4, CO6)

- (i) Concentrations
- (ii) Velocities
- (iii) Fluxes

(b) Define Fick's law. What is mass transfer coefficient ? Derive general mass diffusion equation in a stationary media.

(CO1, CO5)

(c) Define mass transfer by convection. Define diffusional resistance and hence derive the expression for steady state diffusion through a plane membrane.

(CO3, CO4)

Roll No.

TPE-404

B. TECH. (PE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022

APPLIED SEDIMENTOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What do you understand by weathering ? Discuss the different types of chemical weathering. (CO1)
- (b) Describe the term "diagenesis" and explain how it affects the reservoir quality. (CO1, CO3)
- (c) Discuss the role of fluid density in determining fluid transport. Explain different types of fluid flows with the help of suitable diagrams. (CO3)
2. (a) Describe in detail about the agents involved in the transportation of sediments. (CO4)

P. T. O.

(2)

- (b) Explain how the distance of sediment transportation affects size, roundness and sphericity of sedimentary particles. (CO2)
- (c) Distinguish between porosity and permeability. Discuss their significance in the field of petroleum exploration. (CO5)
3. (a) Explain the various principles of stratigraphy with suitable examples. (CO2)
- (b) Discuss about Secondary sedimentary structures. Add a note on the significance of secondary sedimentary structures. (CO4, CO5)
- (c) Discuss about Primary sedimentary structures. Also highlight the significance of primary sedimentary structures. (CO4, CO5)
4. (a) Explain various types of sedimentary depositional environments. Illustrate the parameters of depositional environments. (CO4)
- (b) Explain the concept of transgression and regression with the help of neat and clean diagrams. (CO5)
- (c) What are various types of sedimentary depositional environments ? Describe the physical and chemical parameters of depositional environments. (CO4, CO5)
5. (a) Explain the following : (CO2, CO4)
- (i) Sedimentary texture
- (ii) Wentworth grade scale
- (b) Give the detailed classification of limestone rocks. (CO6)
- (c) Define sedimentary basin. Discuss the categories of Indian sedimentary basins and also locate them in the roughly drawn sketch map of India. (CO5)

Roll No.

TPE-405

B. TECH. (PETROLEUM ENGINEERING)

(FOURTH SEMESTER)

END SEMESTER EXAMINATION, June/July, 2022

PETROLEUM DRILLING ENGINEERING—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List the *six* main systems of a rotary drilling rig. Discuss the component and functions of rotary system of a rotary drilling rig. (CO1)
- (b) What is kick ? List five major causes of kicks. What is the typical sequence of events that may be observed if a kick is taken during a drilling ? (CO1)

P. T. O.

(c) Well control procedures vary slightly from rig to rig, and company to company, but four simultaneous operations are normally considered. List the main operations of well control. Discuss the Driller's method of kick control. (CO1)

2. (a) List and describe the ten functions of drilling fluid. (CO3)

(b) What is yield point of a drilling fluid ? What is the difference between yield point and gel strength. Describe three measures to lower the yield point of a drilling fluid. (CO3)

(c) What is Bingham plastic model of fluid behaviour ? Can the fluid behaviour be accurately modelled by the Newtonian, Bingham, plastic or power law model. (CO3)

3. (a) What is the purpose of casing ? List and describe the different types of casing used in rotary drilling. (CO4)

(b) What is the purpose of cementing ? List the different cementing procedure. Explain primary cementing and squeeze cementing. (CO4)

(c) The following problems are encountered during a cementing operation. Explain the causes of the problem and correlation to be applied : (CO4)

(i) Cannot receive fluid from the rig.

(ii) Cannot pump out of displacement tank

(iii) No cement returns to surface

(iv) Sudden pressure drop while displacing

4. (a) An intermediate casing string (Fig. 1) will be cemented as shown. Calculate the hydrostatic pressure at 12,000 ft. Convert the pressure at 12,000 ft to an equivalent mud weight and determine if it will exceed the fracture gradient of 14.2 lb/gal. (CO5)

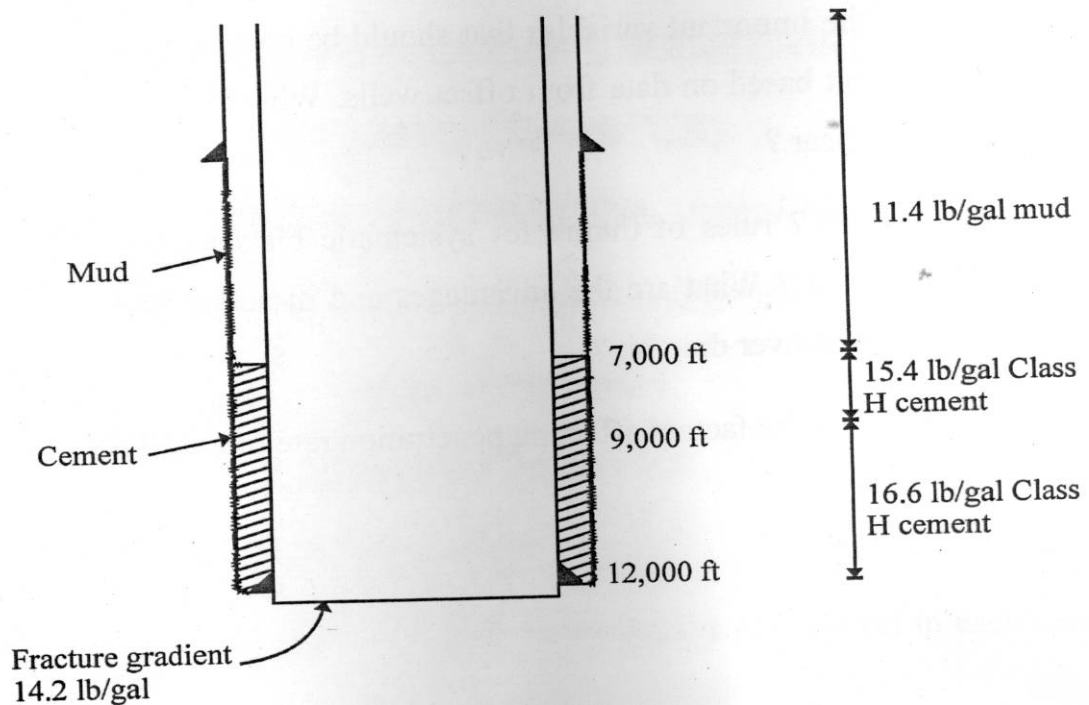


Fig. 1

- (b) Calculate slip velocity using the following information : (CO5)

Given : Drilling Lithology : Shale @ 2.4 g/cc

Particle size : 1/2" by 1/2"

Viscosity : 55 cps

Annular Velocity : 150 ft/min

Gravity : 32.2 ft/sec²

Flow Regime : Laminar

P. T. O.

(c) What are the different types of flow regimes in hydraulics of drilling fluid ? Turbulent flow has greater cuttings cleaning and carrying capacity than laminar flow. Why is it not normally the flow regime of choice ? (CO5)

5. (a) List the important variables that should be considered when selecting a drill bit based on data from offset wells. What are the factors affecting tooth wear ? (CO6)

(b) List the 7 rules of thumb for systematic bit selection. What is roller cutter bit ? What are the advantages and disadvantages of using roller cutter bit over drag bit ? (CO6)

(c) What are the factors affecting penetration rate of a drill bit ? Discuss.

(CO6)

Roll No.

TPE-413

**B. TECH. (PE) (FOURTH SEMESTER)
END SEMESTER EXAMINATION, June/July, 2022**

MUD LOGGING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Mud logging unit ? Describe different types of Mud logging unit. Write the name of sensors available on different Mud logging units. (CO1)

(b) How Mud Logging unit is used at well site ? Describe its functions. (CO2)

(c) Write a note on gas hydrate. (CO4)

2. (a) Describe Flame Ionisation Detector (FID), Stand pipe pressure and Pit volume sensors in Mud Logging Unit. (CO2)

P. T. O.

(2)

- (b) Describe the role of petroleum engineers at well site where mud logging unit has been deployed. (CO1)
- (c) Write a note on use of mud logging system for safety, efficiency and formation evaluation. (CO1)
- 3. (a) Write any *two* best institutions of Petroleum Engg. in India.
- (b) What is Lag time ? How is it calculated ? (CO1)
- (c) Describe shale shaker, degasser, desander and desilter. (CO5)
- 4. (a) How the cutting samples are collected and packed at well site ? (CO3)
- (b) How master charts and well completion reports are prepared ? (CO4)
- (c) Describe how ditch samples are described. (CO3)
- 5. (a) What is mud ? Describe various functions of mud. (CO6)
- (b) Describe temperature, torque and hook load sensors. (CO6)
- (c) Describe any *four* sensors of mud logging unit. (CO2)